Chapter 5 Homework

Part A Multiple choice (Select the one that is best in each case. 1 point/question)

1. Which of the following is a statement of the first law of thermodynamics?
A) A negative ΔH corresponds to an exothermic process.
B) Energy lost by the system must be gained by the surroundings.
C) $\Delta E = E_{\text{final}} - E_{\text{initial}}$
D) $E_k = \frac{1}{2}mv^2$
E) 1 cal = 4.184 J (exactly)

2. For a particular process $q = -10$ kJ and $w = 25$ kJ. Which of the following statements is true?
A) The system does work on the surroundings.
B) Heat flows from the surroundings to the system.
C) $\Delta E = 15$ kJ
D) All of these are true.
E) None of these is true.

3. Which one of the following is an endothermic process?
A) ice melting
B) water freezing
C) boiling soup
D) Hydrochloric acid and barium hydroxide are mixed at 25 °C: the temperature increases.
E) Both A and C

4. Which statement is true of a process in which 1 mol of a gas is expanded from state *A* to state *B*?
A) The final volume of the gas will depend on the path taken.
B) The amount of work done in the process must be the same, regardless of the path.
C) When the gas expands from state *A* to state *B*, the surroundings are doing work on the system.
D) The amount of heat released in the process will depend on the path taken.
E) It is not possible to have more than one path for a change of state.

5. Given the data in the table below, $\Delta H°_{rxn}$ for the reaction

$$C_2H_5OH(l) + O_2(g) \rightarrow CH_3CO_2H(l) + H_2O(l)$$

is ________ kJ.

Substance	$\Delta H°_f$ (kJ/mol)
C_2H_4 (g)	52.3
C_2H_5OH (l)	–277.7
CH_3CO_2H (l)	–484.5
H_2O (l)	–285.8

A) –79.0
B) –1048.0
C) 492.6
D) – 492.6
E) The value of $\Delta H°_f$ of $O_2(g)$ is required for the calculation.

6. The molar heat capacity of an unknown substance is 92.1 J/mol-K. If the unknown has a molar mass of 118 g/mol, what is the specific heat (J/g-K) of this substance?
A) 1.28 B) – 92.1 C) 1.09×10^4 D) 0.781 E) 92.1

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7. For the process $H_2O(l) \rightarrow H_2O(g)$ at 298 K, 1.0 atm, ΔH is more positive than ΔE by 2.5 kJ/mol. This quantity of energy can be considered to be
A) the heat flow required to maintain a constant temperature.
B) the work done in pushing back the atmosphere.
C) the value of ΔH itself.
D) the difference in the H—O bond energy in $H_2O(l)$ compared to $H_2O(g)$.
E) none of these

8. The specific heat of liquid bromine is 0.226 J/g-K. How much heat (J) is required to raise the temperature of 20.0 mL of bromine from 25.00 °C to 27.30 °C? The density of liquid bromine: 3.12 g/mL.
A) 5.20 B) 16.2 C) 300. D) 32.4 E) 10.4

9. Which one of the following statements is true?
A) The enthalpy change of a reaction is the reciprocal of the ΔH of the reverse reaction.
B) ΔH is the value of q measured under conditions of constant volume.
C) Enthalpy is an intensive property.
D) The enthalpy change for a reaction is independent of the state of the reactants and products.
E) Enthalpy is a state function.

10. For which one of the following reactions is ΔH°_{rxn} equal to the standard enthalpy of formation of the product?
A) $2\ C(graphite) + 2\ H_2(g) \rightarrow C_2H_4(g)$
B) $N_2(g) + O_3(g) \rightarrow N_2O_3(g)$
C) $N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3(g)$
D) $12\ C(graphite) + 11\ H_2(g) + 11\ O(g) \rightarrow C_{12}H_{22}O_{11}(s)$
E) $CO(g) + (1/2)\ O_2(g) \rightarrow CO_2(g)$

11. The value of ΔH° for the reaction below is 186 kJ.

$$2\ HCl(g) \rightarrow H_2(g) + Cl_2(g)$$

The value of ΔH°_f for $HCl(g)$ is ________ kJ/mol.
A) -3.72×10^2 B) 186 C) – 93.0 D) – 186 E) 93.0

12. A 50.0-g sample of liquid water at 25.0 °C is mixed with 23.0 g of water at 79.0 °C. The final temperature of the water is ________ °C.
A) 123 B) 27.3 C) 52.0 D) 231 E) 42.0

13. Which substance in the table undergoes the greatest temperature change when the same mass of each substance absorbs the same quantity of heat?

Elements		Compounds	
Substance	Specific Heat (J/g-K)	Substance	Specific Heat (J/g-K)
Al(*s*)	0.90	$H_2O(l)$	4.18
Fe(*s*)	0.45	$CO_2(g)$	0.84
Hg(*l*)	0.14	$CaCO_3(s)$	0.82

A) Al(*s*) B) $H_2O(l)$ C) Hg(*l*) D) Fe(*s*) E) $CaCO_3(s)$

14. The combustion of titanium with oxygen produces titanium dioxide:

$$Ti(s) + O_2(g) \rightarrow TiO_2(s)$$

When 0.610 g of titanium is combusted in a bomb calorimeter, the temperature of the calorimeter increases from 25.00 °C to 50.50 °C. In a separate experiment, the heat capacity of the calorimeter is measured to be 9.84 kJ/K. The heat of reaction for the combustion of a mole of Ti in this calorimeter is ________ kJ/mol.

A) -1.97×10^4 B) –251 C) –19690 D) –19.7 E) 1.97×10^4

15. If a student performs an endothermic reaction in a calorimeter, how does the calculated value of ΔH differ from the actual value if the heat exchanged with the calorimeter is not taken into account?

A) ΔH_{calc} would be more negative because the calorimeter always absorbs heat from the reaction.
B) ΔH_{calc} would be less negative because the calorimeter would absorb heat from the reaction.
C) ΔH_{calc} would be more positive because the reaction absorbs heat from the calorimeter.
D) ΔH_{calc} would be less positive because the reaction absorbs heat from the calorimeter.
E) ΔH_{clac} would equal the actual value because the calorimeter does not absorb heat.

Part B Short questions (15 points)

16. (4 points) Assume that the following reaction occurs at constant pressure:

$2\ Al(s) + 3\ Cl_2(g) \rightarrow 2\ AlCl_3(s)$

(a) If you are given ΔH for the reaction, what additional information do you need to determine ΔE for the process?
(b) Which quantity is larger for this reaction? Explain.

17. (5 points) Consider the following reaction:
$2\ Mg(s) + O_2(g) \rightarrow 2\ MgO(s) \quad \Delta H = -1204$ kJ
(a) Is this reaction exothermic or endothermic?
(b) Calculate the amount of heat transferred when 3.59 g of Mg(s) reacts at constant pressure.
(c) How many grams of MgO are produced during an enthalpy change of –234 kJ ?

18. (3 points) Calculate the standard enthalpy of formation of gaseous diborane (B_2H_6) using the thermal information below.

(1) $4\ B\ (s) + 3O_2\ (g) \rightarrow 2\ B_2O_3(s)\ \triangle H^0 = -2509.1$ kJ
(2) $2H_2\ (g) + O_2\ (g) \rightarrow 2\ H_2O(l)\ \triangle H^0 = -571.7$ kJ
(3) $B_2H_6\ (g) + 3\ O_2\ (g) \rightarrow B_2O_3\ (s) + 3\ H_2O(l)\quad \triangle H^0 = -2147.5$ kJ

19. (3 points) Consider the reaction

$$2\ HCl(aq) + Ba(OH)_2(aq) \rightarrow BaCl_2(aq) + 2\ H_2O(l)\ \ \Delta H = -118 \text{ kJ}$$

Calculate the heat when 100.0 mL of 0.500 M HCl is mixed with 300.0 mL of 0.100 M $Ba(OH)_2$. Assuming that the temperature of both solutions was initially 25.0 °C and that the final mixture has a mass of 400.0 g and a specific heat capacity of 4.18 J/g-°C, calculate the final temperature of the mixture.

Chapter 5 Homework Answer Sheet

Name: **Student ID:** **Instructor:** **Score:**

Part A Multiple choice (15 points)

1-5______________

6-10_____________

11-15____________

Part B Short questions (15 points)